

## 6.12 Case Study: MVP for a Canteen Pre-Order System

ایک کالج میں طلبہ کینٹین کی لمبی قطاروں میں انتظار کرتے تھے۔

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 13 of 16

### 6.12 Case Study: MVP for a Canteen Pre-Order System

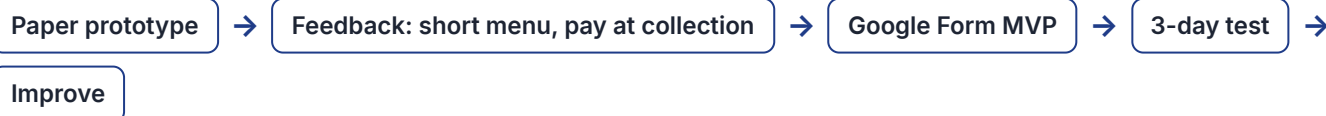
#### LEARN IT

**Problem:** In a college, students wait in long lines at the canteen. Sometimes snacks finish before students reach the counter. A group wants to test a canteen pre-order idea.

**Goal:** Their aim is **NOT** to create a complete app immediately. Their aim is to test **whether students will place orders before break** and **whether the canteen can manage these orders**.

#### The approach

1. The team creates a **paper prototype** showing three steps: **select item, enter name, collect order**.
2. **Feedback** from the prototype suggests the **menu should be short** and **payment should be at collection time**.
3. For the actual MVP, the team uses a **Google Form**. Students select **one of five items**.
4. The canteen prepares these pre-orders. This MVP is tested for **three days only** to learn quickly.



**Table 6.9: MVP plan for canteen pre-order system**

| MVP Element    | Decision                                            |
|----------------|-----------------------------------------------------|
| Features       | Menu list, student name, class, item, and quantity. |
| Cost & Tools   | Free Google Form or printed order slips.            |
| Marketing      | Class announcement and WhatsApp group message.      |
| Testing period | Three days.                                         |

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| Feedback method | Ask students if they saved time and would use it again.  |
| Improvement     | Keep only 5 items; close orders 20 minutes before break. |

**Result:** Students liked saving time, but **canteen staff found too many items difficult**. The team improved the MVP by **keeping only five items** and creating a **separate collection line**. This improvement is based on real feedback.

#### WORKED EXAMPLE

**Worked example:** trace the case study through the chapter's ideas.

| Chapter idea                 | Where it appears in the case study                                       |
|------------------------------|--------------------------------------------------------------------------|
| Real problem (6.3)           | Long queues; snacks finish before students reach the counter.            |
| Low-fidelity prototype (6.5) | Paper screens: select item, enter name, collect order.                   |
| Iteration (6.6)              | Feedback led to a short menu and payment at collection.                  |
| MVP (6.8)                    | Google Form with one of five items, used for real orders.                |
| Riskiest assumption (6.10)   | Students will order before break, and the canteen can manage the orders. |
| Beachhead market (6.14)      | One college canteen during break.                                        |

**Why "close orders 20 minutes before break"?** Staff need time to prepare. If orders arrive at the last minute, the pre-order line becomes as slow as the normal line, and the main benefit disappears.

#### REMEMBER IT

Remember the case study as **"5 items, 3 days, 20 minutes"**: five menu items, a three-day test, orders closing twenty minutes before break.

#### COMMON MISTAKE

**Mistake:** Thinking only the students' opinion matters.

**Fix:** The case study tested **two** things: whether students will order **and** whether the canteen can manage the orders. The staff's feedback (too many items) caused the key improvement.

ایک کالج میں طلبہ کینٹین کی لمبی قطاروں میں انتظار کرتے تھے۔ ٹیم نے فوراً ایپ بنانے کے بجائے پہلے کاغذی پروٹوٹائپ بنایا، پھر تین دن کے لیے گوگل فارم کو ایم وی پی کے طور پر آزمایا جس میں پانچ اشیاء تھیں۔ طلبہ کو وقت کی بچت پسند آئی مگر عملے کو زیادہ اشیاء مشکل لگیں، اس لیے صرف پانچ اشیاء رکھی گئیں، الگ قطار بنائی گئی اور بریک سے بیس منٹ پہلے آرڈر بند کیے گئے۔

#### CHECK YOURSELF

1. Which tool was used for the canteen MVP, and for how long was it tested?
2. What two things did the team want to test?
3. What two improvements were made after the result?

Answers: 1. A Google Form, tested for three days. 2. Whether students will place orders before break, and whether the canteen can manage these orders. 3. Keeping only five items and creating a separate collection line (orders also close 20 minutes before break).

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** The canteen case is there to:

- a. Show a full MVP story on a local problem
- b. Teach NAND
- c. Teach OSI
- d. Teach ER only

**MCQ 2.** The problem is:

- a. Queues / food running out
- b. GPUs
- c. Primary keys
- d. Figma auto-layout

**MCQ 3.** A sensible MVP is:

- a. Pre-order a few items for pickup, not a full restaurant OS
- b. Drone delivery in week 1
- c. AI chefs
- d. Blockchain loyalty

**MCQ 4.** The kitchen is a:

- a. User/constraint you must test
- b. A pie
- c. A stack
- d. An OSI hub

**MCQ 5.** If nobody pre-orders, the idea:

- a. Failed the risky assumption — good you learned cheap
- b. Needs more gradients
- c. Needs a CEO
- d. Needs C++

**MCQ 6.** Write the case as:

- a. Problem, users, assumption, MVP, test, result, next
- b. Only the logo
- c. Only careers
- d. Only Python lists

## Extra short questions

1. Retell the case in 5 bullets.
2. Two user groups with different needs.
3. A metric.
4. A next step if it works.

## Extra long question

1. Write the canteen case as a 10-mark structured answer with a tiny flow diagram (order → kitchen → pickup).

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Queue; students+staff+cooks; assume pre-order helps; form/WhatsApp MVP; measure wait/orders; iterate.
2. Students (speed); kitchen (simple tickets, not 40 modifiers).
3. Orders completed; average wait; food wasted.
4. Add payment; then maybe an app.

### Long

1. Headings as above. Diagram three boxes. One risk: kitchen overwhelmed at 10:55. Mitigation: cap orders. Ethics: allergies.

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Next lecture: 6.13 Testing an MVP